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TECHNICAL PRESENTATION

Steganography vs Cryptography: Hidden Security in the Digital World

Cryptography and Network Security (CSA51)

NAME: JEEVAKRISHNA.R
REG NO: 192524131

What is Information Security?

Information security is the practice of protecting information from unauthorized access, modification, disclosure, and destruction.

Two important techniques are:

- Cryptography
- Steganography

Both protect sensitive information but use different approaches.

Image idea:

- Lock (Cryptography)
- Hidden Image (Steganography)

What is Cryptography?

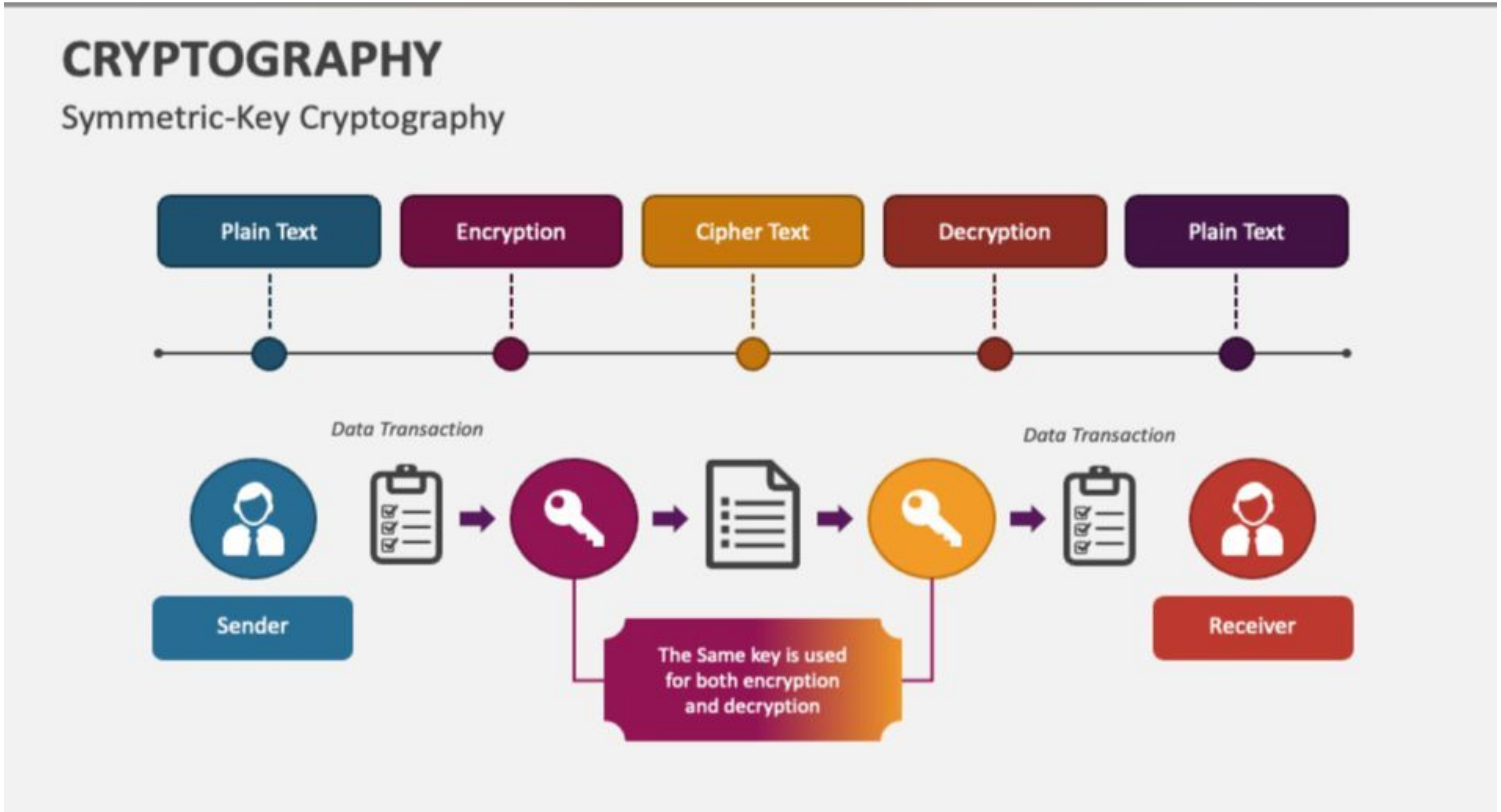
Definition

Cryptography converts readable data (plaintext) into unreadable data (ciphertext) using an encryption algorithm and a key.

Process
Plain
Text
↓
Encryption
↓
Cipher Text
↓
Transmission
↓
Decryption
↓
Original Message

Advantages

- Confidentiality
- Authentication
- Integrity
- Non-repudiation


CRYPTOGRAPHY
Symmetric-Key Cryptography
Plain Text
Encryption
Cipher Text
Decryption
Plain Text
Data Transaction
Data Transaction
Sender
The Same key is used for both encryption and decryption
Receiver

What is Steganography?

Definition

Steganography is the art of hiding secret information inside another digital object so that no one knows a secret message exists.

Carrier can be

- **Image**
- **Audio**
- **Video**
- **Text**

Process

Secret Message

↓

Hide inside Image

↓

Stego Image

↓

Receive Image

↓

Extract Secret Message

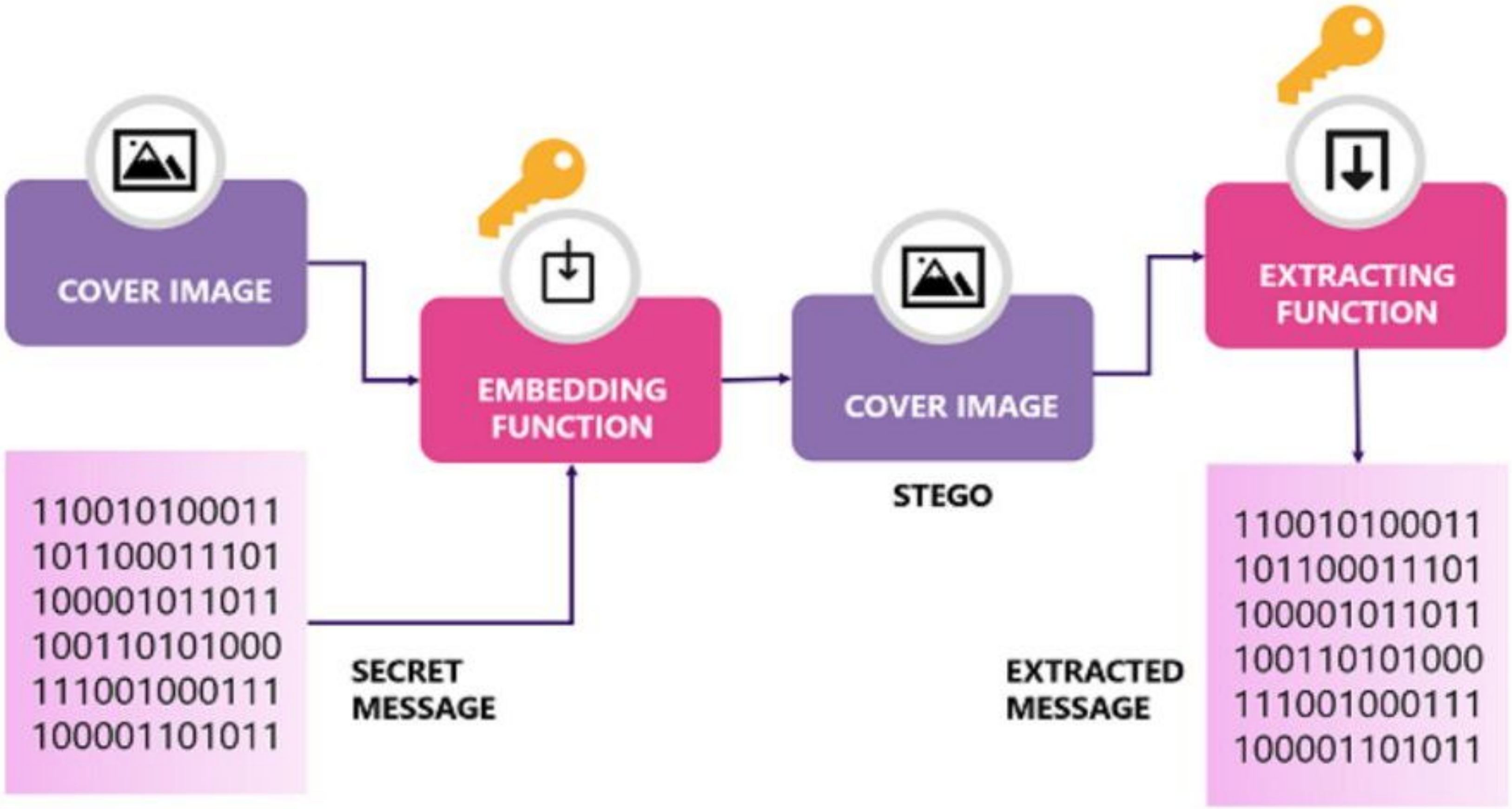


Fig 1: Functional diagram of a steganography system which works on images

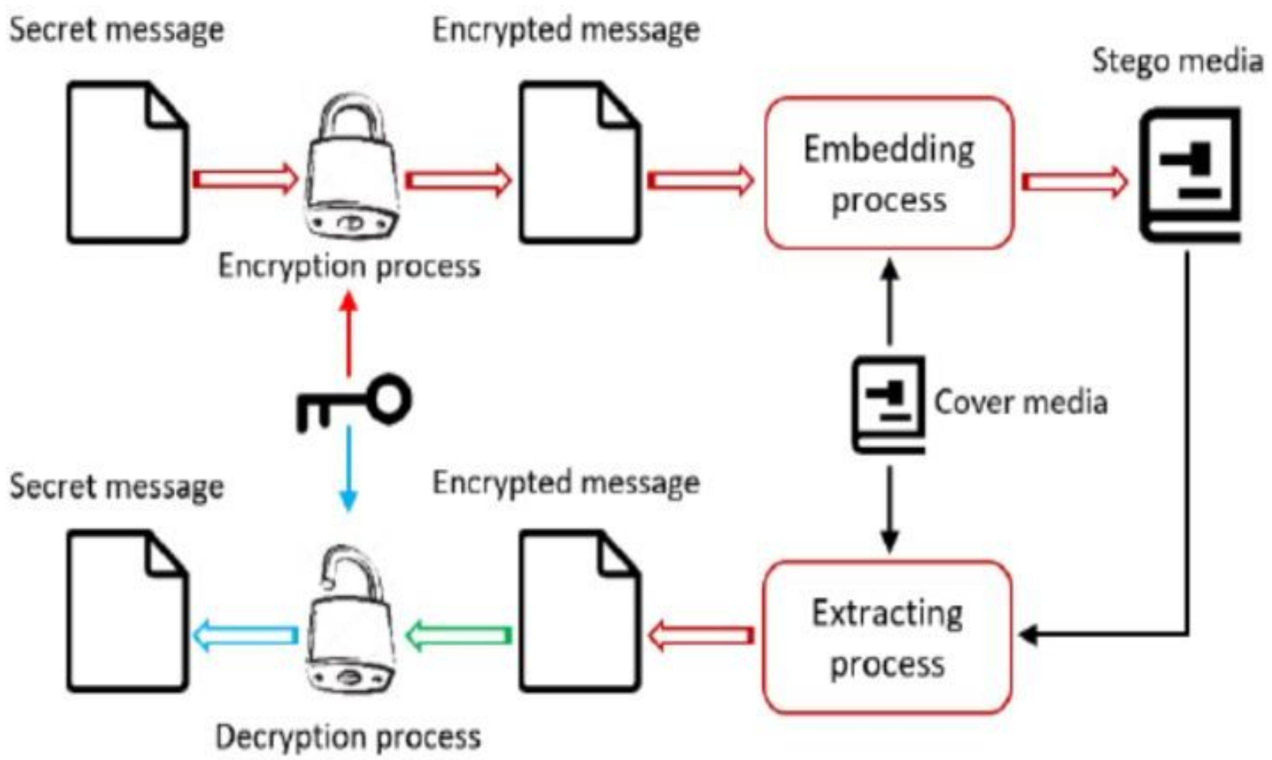
Cryptography vs Steganography

Cryptography	Steganography
Encrypts data	Hides data
Everyone knows encrypted data exists	Nobody knows secret data exists
Produces ciphertext	Produces normal-looking image
Focuses on confidentiality	Focuses on secrecy
Easy to detect encrypted data	Difficult to detect hidden data

Combining Both Techniques

Best security is achieved by combining both.

Message
↓
Encrypt
↓
Cipher Text
↓
Hide inside Image
↓
Stego Image
↓
Extract
↓
Decrypt
↓
Original Message



This provides two layers of protection.

Proposed System

Flowchart

User Input
↓
Encrypt Message
↓
Cipher Text
↓
Hide in Image (LSB)
↓
Stego Image
↓
Receive Image
↓
Extract Cipher Text
↓
Decrypt
↓
Original Message

Experimental Setup

Hardware

- Laptop
- 8 GB RAM

Software

- Python 3.12
- VS Code
- OpenCV
- Pillow
- NumPy

Algorithm

- Caesar Cipher (or Vigenère Cipher)
- LSB Image Steganography

Applications

- Military Communication
- Banking
- Medical Records
- Digital Watermarking
- Cybersecurity
- Cloud Storage
- Intelligence Agencies

Advantages and Limitations

Cryptography

Advantages

- Strong encryption
- Authentication
- Integrity

Limitations

- Ciphertext attracts attackers
- Key management required

Steganography

Advantages

- Hidden communication
- Difficult to detect
- Extra secrecy

Limitations

- Limited storage capacity
- Can be destroyed by image compression

Conclusion

- Cryptography protects message content through encryption.
- Steganography conceals the existence of the message.
- Combining both techniques provides stronger security than using either alone.
- The practical demonstration shows secure communication with two layers of protection.
- These methods are widely used in modern cybersecurity applications.

References

- William Stallings – *Cryptography and Network Security*
- Behrouz A. Forouzan – *Cryptography and Network Security*
- NIST Cryptographic Standards
- IEEE Xplore Digital Library
- Springer Research Articles